

NORYL* SE1GFN2 Resin

Sunday, August 19, 2012

General Information

Product Description

PPE+PS blend. 20% Glass reinforced. Non-brominated, non-chlorinated FR system. UL94 V1 and 5VA listing. RTI 110/105/110. Dielectric strength. Dimensional stability. Suitable for E/E applications.

General

Material Status	• Commercial: Active
Availability	• North America
Filler / Reinforcement	• Glass Fiber Reinforcement, 20% Filler by Weight
Additive	• Flame Retardant
Features	• Bromine Free • Chlorine Free • Flame Retardant • Good Dimensional Stability
Uses	• Electrical/Electronic Applications
Processing Method	• Injection Molding
Multi-Point Data	• Coefficient of Thermal Expansion vs. Temperature (ASTM E831)
	• Elastic Modulus vs Temperature (ASTM D4065)
	• Flexural DMA (ASTM D4065)
	• Specific Heat vs. Temperature (ASTM D3417)
	• Tensile Creep (ASTM D2990)
	• Tensile Fatigue
	• Tensile Stress vs. Strain (ASTM D638)
	• Viscosity vs. Shear Rate (ASTM D3835)

ASTM & ISO Properties ¹

Physical	Nominal Value	Unit	Test Method
Specific Gravity	1.23	g/cm ³	ASTM D792
Molding Shrinkage - Flow (3.20 mm)	0.20 to 0.50	%	Internal Method
Water Absorption (24 hr)	0.060	%	ASTM D570
Mechanical	Nominal Value	Unit	Test Method
Tensile Strength ² (Break)	107	MPa	ASTM D638
Tensile Elongation ² (Break)	5.0	%	ASTM D638
Flexural Modulus ³ (100 mm Span)	5720	MPa	ASTM D790
Flexural Strength ³ (Yield, 100 mm Span)	152	MPa	ASTM D790
Impact	Nominal Value	Unit	Test Method
Notched Izod Impact			ASTM D256
-40°C	96	J/m	
23°C	110	J/m	
Hardness	Nominal Value	Unit	Test Method
Rockwell Hardness (L-Scale)	106		ASTM D785

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SABIC Innovative Plastics - Polyphenylene Ether + PS

Thermal	Nominal Value	Unit	Test Method
Deflection Temperature Under Load 0.45 MPa, Unannealed, 6.40 mm	138	°C	ASTM D648
Deflection Temperature Under Load 1.8 MPa, Unannealed, 6.40 mm	132	°C	ASTM D648
CLTE - Flow (-40 to 95°C)	0.000036	cm/cm/°C	ASTM E831
RTI Elec	110	°C	UL 746
RTI Imp	105	°C	UL 746
RTI Str	110	°C	UL 746
Electrical	Nominal Value	Unit	Test Method
Dielectric Strength (3.20 mm, in Oil)	24	kV/mm	ASTM D149
Dielectric Constant			ASTM D150
50 Hz	2.98		
60 Hz	2.98		
Dissipation Factor			ASTM D150
50 Hz	0.0016		
60 Hz	0.0016		
Arc Resistance ⁴	PLC 7		ASTM D495
Comparative Tracking Index (CTI)	PLC 1		UL 746
High Amp Arc Ignition (HAI)	PLC 1		UL 746
High Voltage Arc Tracking Rate (HVTR)	PLC 4		UL 746
Hot-wire Ignition (HWI)	PLC 0		UL 746
Flammability	Nominal Value	Unit	Test Method
Flame Rating			UL 94
5.99 mm	V-0		
1.47 mm	V-1		
2.50 mm	5VA		
Oxygen Index	31	%	ASTM D2863

Processing Information

Injection	Nominal Value	Unit
Drying Temperature	110 to 121	°C
Drying Time	3.0 to 4.0	hr
Drying Time, Maximum	8.0	hr
Suggested Max Moisture	0.020	%
Suggested Shot Size	30 to 70	%
Rear Temperature	266 to 316	°C
Middle Temperature	277 to 321	°C
Front Temperature	288 to 327	°C
Nozzle Temperature	299 to 327	°C
Processing (Melt) Temp	299 to 327	°C
Mold Temperature	82.2 to 110	°C
Back Pressure	0.345 to 0.689	MPa

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SABIC Innovative Plastics - Polyphenylene Ether + PS

Injection	Nominal Value	Unit
Screw Speed	20 to 100	rpm

Notes

¹ Typical properties: these are not to be construed as specifications.

² Type I, 5.0 mm/min

³ 2.6 mm/min

⁴ Tungsten Electrode

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